DIPLOMA PROGRAMME PRACTICE · HL · SET A

Biology / Paper 1B

Question booklet · 35 marks · 120 minutes across 1A and 1B

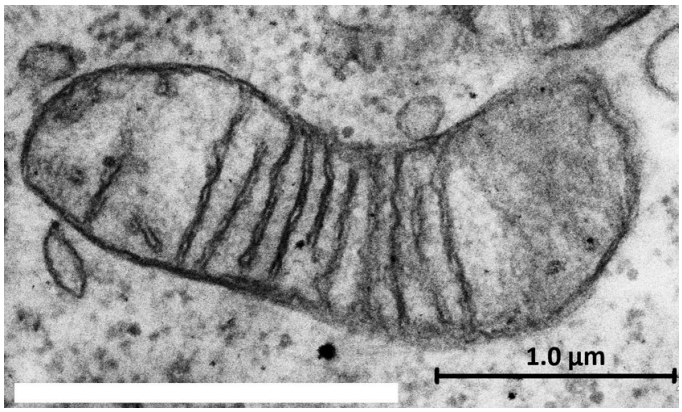
Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams.

QUESTION 1 [8]

Interpreting an electron micrograph

The electron micrograph includes a scale bar and one large organelle in section. A separate study counts inner-membrane profiles intersected by equal-length test lines placed at random in two muscle samples.

Tissue	Total test-line length / µm	Intersections
X	40	48
Y	60	90



Electron micrograph with 1.0 µm bar. Supplied HL Paper 1B model answers, p. 2.

(a) Identify the large organelle and give one visible feature supporting the identification. **[2]**

(b) Estimate its maximum visible length relative to the scale bar and give the approximate length. **[2]**

(c) Compare the two normalized intersection counts. Why does the larger raw count alone not establish a difference in membrane density? **[2]**

Paper 1B · Question 1 (continued)

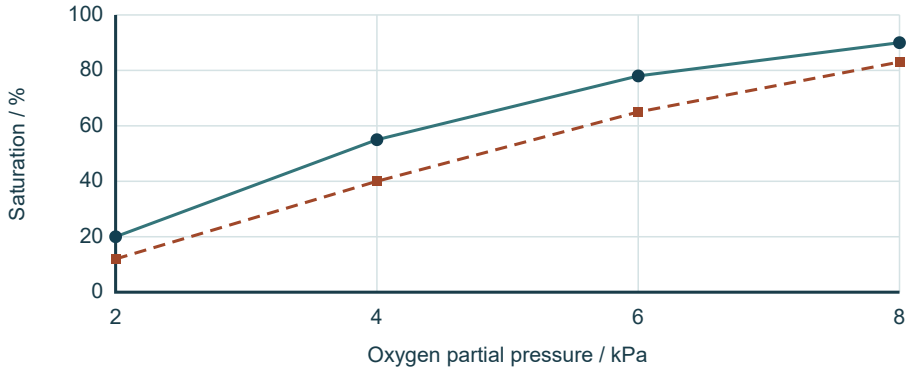
(d) Explain why the count alone does not establish a higher ATP synthesis rate in the intact tissue. **[2]**

QUESTION 2 **[9]**

Oxygen delivery under two conditions

The table gives measured saturation of human haemoglobin at two pH values. For this model take oxygen capacity as 200 cm³ O_2 per dm³ fully saturated blood and neglect dissolved oxygen. Arterial blood begins at 97% saturation.

Tissue oxygen partial pressure / kPa	2	4	6	8
Saturation at pH 7.4 / %	20	55	78	90
Saturation at pH 7.2 / %	12	40	65	83



Circles and solid line: pH 7.4. Squares and dashed line: pH 7.2. Lines connect measured values.

(a) At 4 kPa calculate O_2 unloaded per dm³ blood at each tissue pH. **[3]**

(b) Relate the change to respiration in an active tissue. **[3]**

(c) Blood flow in a muscle is 0.60 dm³ min^{-1} at the lower pH. Estimate delivery to the tissue per minute and give two assumptions behind applying the calculation. **[3]**

QUESTION 3 [9]

Correcting a respirometer

Equal-mass germinating seeds are in a respirometer with a CO_2 absorbent. A matched apparatus containing inert beads corrects for temperature and pressure changes. Displacement toward the chamber is positive. Calibration is 0.40 mm³ gas per mm displacement. Seed mass is 1.60 g.

Time / min	0	4	8	12
Seeds / mm	0	5	12	18
Beads / mm	0	1	2	3

(a) Calculate corrected oxygen uptake per gram per minute over 12 min. **[3]**

(b) Explain the role of the absorbent and the bead apparatus. **[3]**

(c) A second set at 30 °C has a higher corrected rate. State a suitable experimental improvement and explain why this result does not prove a higher ATP yield per oxygen molecule. **[3]**

QUESTION 4 [9]

A test cross with unequal offspring counts

A double heterozygote is crossed with a double recessive. Each recorded progeny class identifies the gamete contributed by the heterozygote. All progeny are sampled at the same stage.

Gamete	MN	mn	Mn	mN
Offspring	184	196	66	54

(a) Identify the likely parental arrangement and estimate the recombination frequency. **[3]**

(b) What counts would independent assortment predict, assuming equal survival? Use the observed pattern to assess that model. **[3]**

(c) Explain how crossing over makes the less common classes and give one reason the observed percentage need not equal the exact fraction of meiotic cells with any crossover. **[3]**